🧠 Entrepreneurship Bootcamp — Complete Study Guide

Cheatsheet by Makram ElJamal for Entrepreneurship Bootcamp · Spring 2026

📋 PEARSON CRITERIA → WHERE TO STUDY IT P = Pass · M = Merit · D = Distinction

1 · Entrepreneurial Mindset & Approaches

Criterion	Study in this guide
P1 — key skills + entrepreneurial mindset to launch a start-up	L1: Innovation & Entrepreneurship; 3Ms; Growth Mindset; Agility; High-Performance Teams
P2 — entrepreneurship approaches (agile, lean, social)	L3: Entrepreneurship Approaches
M1 — PEST analysis; identify external O&T; justify pursuing an opportunity	L2: PEST + SWOT (the O & T)
D1 — justify the most suitable approach (integrate O&T, market, resources, team, goals)	L3 approaches + L2 PEST/SWOT (e.g. Lean for an uncertain market + limited resources)

2 · Design Thinking & Prototyping

Criterion	Study in this guide
P3 — DT techniques to identify market gaps & challenges	L1 DT process / HCD / Human Hacks; L2 Observe·Engage·Immerse; L3 Problem Statement
P4 — DT to propose a solution + build a prototype	L5 HMW & Ideation; L6 Prototype Types
M2 — multiple prototype iterations from validated feedback; problem–solution fit	L6 Validation Engine; L7 Feedback Extraction & Methods
D2 — critically evaluate validation/testing (problem validation, problem–solution & product–market fit)	L6 Double Diamond; L7 RAT + Iterate/Pivot/Kill

3 · Marketing, Market Size & Finance

Criterion	Study in this guide
P5 — analyse target market + estimate market size	L5 Markets / Segmentation; TAM/SAM/SOM; Diffusion of Innovation
P6 — competitor analysis + clear value proposition	L6 Competitor Analysis / USP / Perceptual Map; L5 Value Proposition Canvas
P7 — cost structures, revenue streams/model, financial projections	L8 Cost, Revenue & Finance Basics
M3 — marketing mix → strategy communicating the VP & competitive advantage	L7 4Ps → 7Ps
M4 — forecasted budget + key financial statements	L8 Budget; Finance Deep-Dive: Income Statement, Balance Sheet & Cash Flow layouts
D3 — feasibility & viability via financial ratios & analysis; justify improvements	Finance Deep-Dive: NPM/GPM/ROI/Current Ratio, Breakeven + all worked examples

4 · Business Model Canvas & Pitch

Criterion	Study in this guide
P8 — craft & describe a Business Model Canvas	L4 Business Model Canvas (9 blocks)
P9 — presentation & persuasion to pitch a business idea	L4 Jargon Assassin ("Speak Human"); L5 Communication & The Pause
M5 — compelling pitch applying entrepreneurial knowledge to investors	Whole guide — lead with VP, market size, BMC & financial viability
D4 — critically evaluate the BMC; improve desirability / feasibility / viability	L4 BMC + Common Mistakes; "best single BMC change" in Exam Strategy

📊 BUSINESS TRACK — Zaid Maaytah · Lectures 1–9

L1 · Innovation & Entrepreneurship

Innovation = creating **and** implementing new ideas that generate value (economic, social, or technological). It can be a new product, service, process, or business model.

Key idea: innovation ≠ invention. An invention is a new thing; an innovation is a new thing that **creates value** and is actually put to use. If it doesn't create value, it isn't innovation.

Entrepreneurship = identifying opportunities, mobilizing resources, and creating value through new ventures. It turns innovative ideas into sustainable businesses, and is the **4th factor of production** alongside land, labor, and capital — the function that organizes the other three.

Innovation Sweet Spot — good innovation sits where 3 circles overlap:

- **Desirability** — do people actually want it? (the human lens)
- **Feasibility** — can it technically be built/delivered?
- **Viability** — can the business make money and sustain itself?

⚡ All 3 must overlap. Missing one: desirable+feasible but not viable = no business; desirable+viable but not feasible = can't build it.

L1 · Design Thinking Process & the DT↔Business Link

Design Thinking (Stanford d.school) — a 5-stage process that starts with **people and their needs**, not with technology or a business plan:

Empathize → Define → Ideate → Prototype → Test

- **Empathize** — understand the user deeply
- **Define** — frame the real problem
- **Ideate** — generate many possible solutions
- **Prototype** — build cheap, rough versions
- **Test** — put them in front of users and learn

Airbnb example: bookings were flat → they discovered listing photos were poor → sent professional photographers → bookings rose. They solved a **user problem**, not a tech problem.

⚡ Design Thinking decides **WHAT** to build (the right problem/solution); Entrepreneurship decides **HOW** to deliver and scale it. Both rely on experimentation, iteration, and learning from users.

L1 · 3Ms, Growth Mindset, Agility & Teams

The 3Ms of entrepreneurship — three things every founder must manage together:

- **Mind** — your mindset; sets direction, resilience, adaptability
- **Market** — the map; where value exists and who needs what you build
- **Money** — the fuel; without it you can't go far, so manage it wisely

Growth Mindset = belief that ability grows through effort; failure is treated as **learning**. Core traits: Curiosity, Vision, Risk Tolerance, Resilience, Adaptability, Persistence.

Business Agility = ability to adapt quickly to changes in markets, tech, and customer needs. Built on rapid experimentation, continuous learning, iterative development, and flexible decision-making.

High-Performance Teams aren't just talented individuals — they collaborate effectively. **CATme** (Comprehensive Assessment of Team Member Effectiveness) is a peer-feedback system that raises accountability and exposes contribution imbalances.

L2 · PEST Analysis

PEST = a framework for scanning the external **macro-environment** — the big forces outside the company that you can't control but must respond to. Used when evaluating a new market or designing a business model.

Political	Government policy, regulation, political stability, support for entrepreneurship, licenses, public funding
Economic	Inflation, interest rates, (un)employment, consumer purchasing power, availability of capital/credit
Social	Cultural values, demographics & population trends, education/literacy, lifestyle and social trends
Technological	Internet/mobile access, infrastructure & connectivity, emerging technologies, digital tools/platforms

⚡ PEST ≠ PESTEL. PESTEL adds **Environmental + Legal** (taught in the Design track). PEST feeds the **Opportunities & Threats** of a SWOT.

L2 · SWOT Analysis

SWOT = a snapshot of where you stand, splitting factors into internal vs external and positive vs negative.

	Positive	Negative
Internal	**Strengths** — what you do well (skilled team, IP)	**Weaknesses** — what you lack (no funding, no brand)
External	**Opportunities** — market trends/gaps you can exploit	**Threats** — competition, regulation, new entrants

How to classify in the exam: ask two questions — (1) is it **inside** the company (something you own/control) or **outside**? (2) is it good or bad? A new competitor lowering prices = outside + bad = **Threat**. A strong technical team = inside + good = **Strength**.

⚡ S+W are **internal**; O+T are **external**. PEST informs the O and T.

L3 · Entrepreneurship Approaches

Lean Startup — eliminate waste by testing before scaling. Runs the **Build → Measure → Learn** loop: build an **MVP** (Minimum Viable Product — the smallest thing that tests your key hypothesis), measure real user behaviour, then decide to **pivot** (change direction) or **persevere**. *Dropbox tested demand with a demo video before writing code.*

Agile — iterative development in short **sprints** with continuous delivery and feedback; flexibility to change direction on new information. Originally software, now used broadly.

Tech-led vs Design-led — "we built it because we could" vs "we built it because they needed it." Best innovations are both, but always **start from the human need**.

Social Entrepreneurship — creates social/environmental value using business methods (not pure charity). *Grameen Bank: microfinance for the poor.*

⚡ Lean (validate fast, kill waste) and Agile (adapt in cycles) **complement** each other — they're not opposites. Best fit for limited resources + uncertain market + likely pivot = **Lean**.

L4 · Business Model Canvas — 9 Building Blocks

BMC = a one-page strategic tool to visualise how a business creates, delivers, and captures value. Layout logic: **right** = market/customer, **centre** = value, **left** = operations, **bottom** = money. It asks: is the model **Desirable, Feasible, Viable**?

Customer Segments	Groups you serve (demographic / geographic / behavioural / psychographic). **No defined segment = no customer.**
Value Proposition	The unique value you offer; what problem you solve & why you're different. Ties to Jobs/Pains/Gains.
Channels	How you reach/deliver: sales, communication, distribution channels.
Customer Relationships	Personal assistance / self-service / automated services.
Revenue Streams	How you earn from each segment: product sales, service fees, subscriptions, licensing, advertising.
Key Resources	Assets needed: physical, intellectual (IP), human, financial.
Key Activities	Most important actions: production, marketing/sales, distribution, customer service.
Key Partners	External help: suppliers, manufacturers, distributors, alliances, investors.
Cost Structure	All costs to operate: fixed + variable; economies of scale & scope.

⚡ Common mistakes: too many segments (pick a **beachhead**), a vague value proposition, unrealistic revenue projections, ignoring costs. A "best single change" exam answer usually fixes **desirability + feasibility + viability at once** (e.g. add a key partner that fixes the core failure and unlocks recurring revenue).

L5 · Value Proposition Canvas

Zooms into two BMC blocks (Value Proposition ↔ Customer Segment) to check they actually match.

Customer Profile (right side):

- **Jobs** — tasks the customer wants done (functional, emotional, social)
- **Pains** — frustrations, risks, negative emotions they want to avoid
- **Gains** — positive outcomes they desire

Value Map (left side):

- **Products & Services** — what you offer
- **Pain Relievers** — how you reduce their pains
- **Gain Creators** — extra benefits you produce

Fit = when your Pain Relievers address real Pains and your Gain Creators deliver real Gains.

VP statement template: "[Product] helps [target customer] achieve [key benefit] by [how it solves their pain]." *Uber: a fast, reliable, affordable ride at the tap of a button — no waiting, no negotiation, no cash.*

⚡ Exam cue: "saves users an hour a day" = a **Gain**. "Reduces long delivery waits" = relieving a **Pain**.

L5 · Markets & B2B vs B2C

Market = a system where buyers and sellers exchange goods/services; can be local→global. Types: consumer, financial, business, online.

B2B (to businesses)	B2C (to consumers)
Rational, ROI-driven	Emotional, personal
Long sales cycles	Short cycles
Multiple decision-makers	Usually one buyer

⚡ If the **user isn't the payer** (e.g. procurement buys, employees use) you're in a "system game" with B2B dynamics — design for the decision-maker, not just the user.

L5 · Diffusion of Innovation

How a new product spreads through a population over time — five adopter groups:

Innovators 2.5%	Risk-takers; try things first, before they're proven
Early Adopters 13.5%	Visionaries/opinion leaders; spot the benefit early
Early Majority 34%	Pragmatists; adopt once it's somewhat proven; risk-averse
Late Majority 34%	Skeptics; price-sensitive; adopt late
Laggards 16%	Traditionalists; last to adopt, if ever

⚡ Your **SOM** (first customers) = Innovators + Early Adopters (~16%). They tolerate an unfinished product because they want the benefit now.

L5 · ★ TAM / SAM / SOM — Market Sizing

Three nested estimates of market size, from biggest to most realistic:

TAM — Total Addressable Market: maximum demand if 100% of all possible users adopted. The whole pie.

SAM — Serviceable Available Market: the slice of TAM you can realistically serve given geography, language, constraints.

$\text{SAM} = \text{TAM} \times \text{target-segment }\%$

SOM — Serviceable Obtainable Market: the slice of SAM you can actually capture short-term (1–3 yrs) given competition and capacity. Your "entry wedge."

$\text{SOM} = \text{SAM} \times \text{market-share (conversion) }\%$

Worked example

3,000,000 smartphone users; 60% health-conscious; 25% of those willing & able to pay; 10% Year-1 conversion.

- TAM = **3,000,000**
- SAM = 3,000,000 × 60% × 25% = 1,800,000 × 25% = **450,000**
- SOM = 450,000 × 10% = **45,000**
- Annual revenue from SOM = users × monthly price × 12

⚡ If price rises, fewer people are "willing & able" → the **% shrinks → SAM shrinks → SOM shrinks**. Cap SOM at your operational capacity if conversion would exceed it.

L6 · Competitor Analysis & USP

Competitor types: Direct (same product, same audience), **Indirect** (different product, same customer need), **Emerging** (new entrants with disruption potential).

Benchmarking = comparing your performance/products against rivals. A **Moon Chart** (radar/spider chart) plots several competitors across several attributes at once — great for spotting strengths/weaknesses visually.

Brand Positioning / Perceptual Map = plot brands on 2 key attributes (e.g. Price vs Quality). Empty regions = **market gaps** = under-served opportunities.

USP (Unique Selling Point) = the feature/benefit that makes you stand out. Must be **Unique, Valuable, and Clear**. It lives where "what customers want" meets "what competitors do poorly."

⚡ Benchmarking reveals where you outperform → that gap becomes your USP.

L6 · ★ Pricing Strategies

Cost-Plus	**Price = production cost + markup %.** Simple, guarantees a margin, but ignores what customers will actually pay.
Value-Based	Price on **perceived customer value**, not cost. Needs strong differentiation & knowledge of willingness-to-pay. Used for luxury/branded goods.
Competitive	Price set against rivals (below/at/above). Used when products are similar (low differentiation).
Penetration	**Low** entry price to grab market share fast, raise later. Best for price-sensitive markets with alternatives. Risk: attracts price-chasers.
Skimming	**High** price first (early adopters who'll pay a premium), lower over time to reach the mass market. *New iPhone launch.*
Psychological	Prices that *feel* cheaper/better: charm ($9.99), prestige, bundling, anchoring, artificial scarcity.

⚡ Risk-averse, budget-conscious buyers at launch (e.g. municipalities) → **penetration + strong promotion** to lower barriers and build trust. Premium/skimming would be rejected.

L7 · 4Ps of Marketing (→ 7Ps)

The **Marketing Mix** — the levers you control to take a product to market. Each P must align with the others and with the target customer.

Product	What you sell: features, quality, design, branding, packaging. Core product (the benefit) vs augmented (warranty, support). Lifecycle: Introduction→Growth→Maturity→Decline.
Price	What customers pay; signals value and positions the brand (see pricing strategies).
Place	How & **where/when** customers access it: direct (own store/site), indirect (retailers/distributors), digital (app stores, marketplaces). *Changing delivery windows/routes = a Place decision.*
Promotion	How you communicate value: advertising, PR, social, content, sales promotions. IMC = one consistent message across channels.

7Ps (services) add: **People** (who delivers it), **Process** (how it's delivered), **Physical Evidence** (tangible cues of quality).

⚡ Exam classify: a discount = **Price**; a new meal bundle = **Product**; an influencer campaign = **Promotion**; evening delivery windows = **Place**.

L8 · Cost, Revenue & Finance Basics

Entrepreneurship = creativity + **resource management**. Finance helps you survive, plan, grow, and reduce uncertainty. Most startups fail from **financial mismanagement**, not bad ideas.

Common mistakes: underestimating costs, overestimating revenue, spending before validating demand, poor cash-flow management.

Startup Costs	One-time costs to launch before any revenue: equipment, legal, website, initial inventory, licenses.
Fixed Costs	Don't change with volume: rent, salaries, insurance, software. Present whether you sell 0 or 10,000 units.
Variable Costs	Scale with output: raw materials, packaging, delivery, sales commissions.

$$\text{Total Cost} = \text{Fixed Costs} + (\text{Variable cost/unit} \times \text{Units})$$

Projected Revenue answers WHO buys, WHAT they buy, and what they PAY — grounded in research, not hope. Use a **bottom-up** estimate (start from individual customer behaviour and scale up); avoid the "if we just get 1% of the market" fallacy.

💰 FINANCE DEEP-DIVE — L9 + the Merit/Distinction calculation engine. Master these worked methods.

The Three Financial Statements

Income Statement (P&L)	Revenue − Costs = Profit/Loss **over a period**. Shows profitability.
Balance Sheet	**Assets = Liabilities + Equity** at a single **point in time** (a snapshot).
Cash Flow Statement	Actual cash **in and out** over a period — not the same as profit.

Key terms: Revenue = total income from sales. **COGS** = cost of goods sold (direct cost of producing what you sold). **OpEx** = operating expenses (running the business: marketing, admin, rent).

⚡ Profitable ≠ cash-positive. You can record a sale (profit) before the customer pays, so cash can be negative while profit is positive. **Burn rate** = how fast a startup spends its cash reserves.

Income Statement (P&L) — layout

A "waterfall": start at sales, subtract costs in layers down to the bottom line. Covers a **period** (e.g. a year).

Revenue (Sales)	1,000,000
− Cost of Goods Sold (COGS)	(600,000)
= Gross Profit	**400,000**
− Operating Expenses (OpEx)	(250,000)
= Operating Profit (EBIT)	**150,000**
− Interest	(20,000)
− Tax	(30,000)
= Net Profit	**100,000**

Tips to calculate:

- Work **strictly top-down**; each subtotal feeds the next line.
- Only **direct production** costs go in COGS; running costs (rent, marketing, admin, depreciation) go in OpEx.
- GPM = Gross Profit ÷ Revenue; NPM = Net Profit ÷ Revenue.

⚡ Here GPM = 400,000/1,000,000 = 40%; NPM = 100,000/1,000,000 = 10%.

Balance Sheet — layout

A **snapshot** at one date. Must always balance: **Assets = Liabilities + Equity**.

ASSETS	
Cash	50,000
Accounts receivable	70,000
Inventory	80,000
Current assets	**200,000**
Equipment (non-current)	300,000
Total Assets	**500,000**
LIABILITIES	
Accounts payable	90,000
Short-term loan	60,000
Current liabilities	**150,000**
Long-term loan	150,000
Total Liabilities	**300,000**
EQUITY	200,000
Liabilities + Equity	**500,000**

Tips to calculate:

- If two of Assets / Liabilities / Equity are given, find the third by rearranging **Equity = Assets − Liabilities**.
- **Current** = due/usable within 1 year (cash, receivables, inventory, payables). **Non-current** = long-term (equipment, long-term loans).
- Current Ratio = Current Assets ÷ Current Liabilities = 200,000/150,000 = **1.33**.

⚡ If it doesn't balance, you've missed an item or mis-signed one — recheck before answering.

Cash Flow Statement — layout

Tracks actual **cash in/out** over a period, in three buckets, ending at the change in the bank balance.

OPERATING (day-to-day trading)	
Cash from customers − cash paid out	120,000
INVESTING (buying/selling assets)	
Bought equipment	(300,000)
FINANCING (loans/investment/owners)	
Loan + investor funds received	250,000
Net change in cash	**70,000**
+ Opening cash	30,000
= Closing cash	**100,000**

Tips to calculate:

- Sort every item into **Operating** (sales, salaries, rent), **Investing** (equipment, property), or **Financing** (loans, equity, repayments).
- Net change = sum of all three buckets; **Closing = Opening + Net change**.
- Cash outflows are negative — money received is +, money spent is −.

⚡ A profitable firm can still show negative cash (e.g. big equipment purchase or unpaid invoices) — this statement reveals that, the P&L doesn't.

★ Core Formulas (with meaning)

$$\text{Gross Profit} = \text{Revenue} - \text{COGS}$$

What's left after the direct cost of making the product. Measures production efficiency.

$$\text{Net Profit} = \text{Gross Profit} - \text{OpEx} - \text{interest} - \text{tax}$$

The true "bottom line" after **all** costs.

$$\text{Gross Profit Margin} = \text{Gross Profit} \div \text{Revenue} \times 100$$

$$\text{Net Profit Margin (NPM)} = \text{Net Profit} \div \text{Revenue} \times 100$$

Margins = profit as a **% of revenue**; let you compare companies of different sizes.

$$\text{ROI} = (\text{Gain} - \text{Cost}) \div \text{Cost} \times 100 = \text{Net Profit} \div \text{Invested capital} \times 100$$

Return per dinar invested. Higher = more efficient use of money.

$$\text{Current Ratio} = \text{Current Assets} \div \text{Current Liabilities}$$

Liquidity — can you cover short-term debts? ≥1.5 is a common lender threshold.

★ Breakeven Analysis

The point where **Total Revenue = Total Costs** (zero profit/loss). Answers "how much must we sell to stop losing money?"

$$\text{Contribution Margin (CM)} = \text{Price} - \text{Variable cost/unit}$$

What each unit contributes toward covering fixed costs.

$$\text{Breakeven Units} = \text{Fixed Costs} \div \text{CM}$$

$$\text{Breakeven Revenue} = \text{Breakeven Units} \times \text{Price}$$

Worked example — profit/loss position

Café: Fixed = 40,000 · Price = 4 · Variable = 1.50 · Sold = 12,000 cups.

- CM = 4 − 1.50 = **2.50/cup**
- Total contribution = 12,000 × 2.50 = **30,000**
- Profit = 30,000 − 40,000 = **−10,000 → a loss**
- Breakeven = 40,000 ÷ 2.50 = **16,000 cups** (they sold too few)

⚡ Trap: don't compare **revenue** to fixed costs. Compare **total contribution** (CM × units) to fixed costs.

Worked: Net Profit with adjustments

Sales 1,000,000 · COGS 520,000 · OpEx 280,000. Adjustments: reduce sales 15,000; move 30,000 delivery from COGS→OpEx; add depreciation 20,000 to OpEx.

- Sales = 1,000,000 − 15,000 = **985,000**
- COGS = 520,000 − 30,000 = **490,000**
- OpEx = 280,000 + 30,000 + 20,000 = **330,000**
- Gross Profit = 985,000 − 490,000 = **495,000**
- Net Profit = 495,000 − 330,000 = **165,000**

⚡ Work top-down in order: adjust Sales → adjust COGS → Gross Profit → adjust OpEx → Net Profit.

Worked: Reclassification & margins (Distinction)

Move a cost (e.g. shipping) from **COGS → OpEx**, nothing else changes:

- COGS falls → Gross Profit rises → **GPM increases**
- But total expenses are unchanged → **Net Profit & NPM stay the same**

Reverse (OpEx → COGS): GPM **decreases**, NPM unchanged.

Add **depreciation to OpEx**: NPM **decreases**, GPM unchanged (depreciation isn't in COGS).

⚡ Golden rule: **moving** a cost between lines never changes Net Profit — it only changes where the cut shows up (Gross vs Net). Only **adding/removing** a cost changes Net Profit.

Worked: Price rise vs cost cut

Price 50 · unit COGS 32 · fixed OpEx 120,000 · volume 10,000.
Option 1: raise price to 52. Option 2: cut unit COGS to 30. (volume fixed)

- **Opt 1:** Rev 520,000; NP = (52−32)×10,000 − 120,000 = 80,000; NPM = 80,000/520,000 = **15.4%**
- **Opt 2:** Rev 500,000; NP = (50−30)×10,000 − 120,000 = 80,000; NPM = 80,000/500,000 = **16.0%**

⚡ Same Net Profit on **lower revenue** = higher margin → the **cost cut wins** on NPM.

Worked: Viability thresholds (NPM & ROI)

Lender wants NPM ≥ 12%; investor wants ROI ≥ 20%.
Revenue 2,400,000 · COGS 1,380,000 · OpEx 960,000 · avg invested capital 150,000.

- Net Profit = 2,400,000 − 1,380,000 − 960,000 = **60,000**
- NPM = 60,000 / 2,400,000 = **2.5% → FAILS 12%**
- ROI = 60,000 / 150,000 = **40% → PASSES 20%**

⚡ NPM and ROI measure different things — always test each threshold separately. NPM uses revenue; ROI uses invested capital.

Worked: ROI comparison

Company A: invested 80,000, net profit 12,800. Company B: invested 60,000, net profit 9,000. Who's more efficient?

- A: 12,800 / 80,000 = **16%**
- B: 9,000 / 60,000 = **15%**
- **A wins** — higher return per dinar even if profits look similar.

⚡ Bigger absolute profit ≠ better. Always divide by what was invested (ROI) or by revenue (margin) to compare fairly.

Worked: Comparing two revenue options

Pick the option with the higher Net Profit Margin. Method for each option:

- Add up all **new revenue** (existing + each new stream)
- Add up all **variable costs** (per-unit × units, each stream)
- Add all **fixed** costs (base + extra support)
- Net Profit = Revenue − Variable − Fixed; NPM = NP ÷ Revenue

A subscription stream (high price, low marginal cost) usually beats a low-fee courier stream on margin — compute both, don't guess.

"Additive" means add the new stream on top of the baseline; recompute the whole P&L, then compare margins.

DESIGN THINKING TRACK — Fayzeh Abuhaltam · Lectures 1–7

L1 · Why Startups Fail & Human-Centered Design

Why startups fail: they build for themselves not users; they start with a solution instead of a problem; same-background teams share the same biases and blind spots.

Bad path: Idea → Build → Hope people want it.
Good path: People → Problem → Solution.

Human-Centered Design (HCD) = start with people — their frustrations, needs, behaviour, habits, workarounds, environments. *"Users describe solutions; designers discover needs."*

DT is "**observation without judgment**" — accept you might be wrong about the problem. Your own education/expertise is also a filter that can stop you seeing reality.

L1 · Cognitive Biases in Innovation

Halo Effect	One impressive feature fools you into ignoring the flaws
Similarity (Like-Me)	Assuming users think/behave like you do
Confirmation	Seeking only the info that supports your existing idea
Stereotyping	Over-generalising a whole user group

These biases make you "validate" assumptions instead of genuinely testing them — leading to products built on false premises.

L1 · Human Hacks & the Design Journey

Human Hack = a non-standard **workaround** people create because a system failed to meet a need (infrastructure / design / trust / process failure). **Where there's a workaround, there's an unmet need.** *e.g. propping a laptop on folded paper → unmet need for adjustable height.*

The bootcamp design journey:

Understand → Research → Define → Ideate → Prototype → Test → Evaluate → Pitch

"Business is about the destination. Design is about the maze (discovery). Don't jump over the maze — walk through it."

Decode complaints one level deeper: "wish slides came earlier" → need for **control/predictability**; "hate group projects" → need for **fair workload**; "need a bigger battery" → need for **anxiety reduction**.

Always translate a stated complaint into the underlying need before designing.

L2 · Observe · Engage · Immerse

The problem with self-reporting: people don't do what they *say*, don't do what they *think* they do, and don't do what you *tell* them to do. So watch behaviour, don't trust opinions.

Observe	Watch what people actually do in context — habits, workarounds, friction, hesitation. Don't intervene or suggest.
Engage	Move from watching to asking — but ask **behavioural** questions about real past events, not opinions. Decode signals: checking phone = boredom; avoiding speaking = discomfort.
Immerse	Experience the problem yourself → understand how it **feels**. *e.g. write with your non-dominant hand.*

Observed behaviour > stated preference. Always. Observation shows *what* happens, talking shows *why*, immersion shows *how it feels*.

L2 · Primary vs Secondary Research

Primary (direct): you gather it first-hand — observe in context, behavioural interviews, immersion.

Secondary (indirect / digital shadowing): existing data — reports, statistics, articles. Market research replaces assumptions with evidence.

Every research insight should answer: **who** is the user, **what** do they need, in **what context**?

"You think you understand a problem — but you only saw a symptom." Dig past the symptom to the real need.

L3 · PESTEL & System Thinking

Business sees the **MAP** (where to go); Design sees the **TRAP** (what's blocking people). Human hacks are responses to system failure.

PESTEL = PEST + two more external forces:

P/E/S/T	Political, Economic, Social, Technological (as in PEST)
Environmental	Sustainability, climate, environmental regulation
Legal	Employment law, consumer protection, IP law

System Thinking: your user lives inside a network — you also "sell" to their parents, friends, boss, and culture. **If the user says YES but someone in their environment says NO, the idea is dead.**

Every external pressure creates behaviour — you're solving for the human experiencing those pressures.

L3 · Jobs-to-be-Done & Kano Model

Jobs-to-be-Done: people "hire" a product to do a job. Three job types:

- **Functional** — the practical task (too slow, too costly)
- **Social** — how they appear to others
- **Emotional** — how they feel (anxiety, confidence)

People don't buy a drill; they hire it to make a hole to hang a family photo — to feel like a good parent.

Kano Model (value layers):

Required	Must-have; absence = dealbreaker
Expected	Baseline; users assume it's there (no praise if present, anger if absent)
Desired	Nice-to-have; users prefer it
Unexpected	Delighters; they didn't know they wanted it

⚡ Kano drift: today's **delighter** becomes tomorrow's **expected**, then **required** (e.g. WiFi on planes).

L3 · ★ Problem Statement Formula

[User] struggles with [specific situation] because [root cause], leading to [real consequence].

Good vs Bad
Good: "Busy students on campus skip meals during peak hours because food queues are unpredictable, leading to fatigue and reduced academic performance."
Bad: "Students need better food on campus." → too vague, no root cause, no consequence.

⚡ A strong problem statement names a **specific user + specific situation + root cause + consequence**. Vague statements produce irrelevant solutions.

L4 · Personas & the Beachhead

The myth of "everyone": if your customer is everyone, you have no customer. "Students / People / Users" are **categories, not segments**.

A real user is in a specific situation, doing something specific, with a real struggle. *"If you can't see the moment, you don't understand the problem."*

Persona components: When does the problem happen (situation) · What do they actually do (behaviour) · What are they trying to avoid (pains) · What do they want to feel (emotional job).

Beachhead persona = your one most-focused target = Segment + real human + real moment + job + intensity. Win them first, then expand.

⚡ **Shark bite** (urgent, emotional, act NOW) vs **Mosquito bite** (annoying, ignored, delayed). A beachhead needs a **shark-bite** problem — if it's a mosquito bite, people won't change behaviour to fix it.

L4 · Root Cause & the Jargon Assassin

5 Whys: ask "why?" repeatedly to drill from symptom to **root cause**. Then move from "**What is**" (the problem/cage) to "**What if**" (the solution/exit).

Jargon Assassin: jargon is a **symptom**; unclear thinking is the real problem. Rule: explain your idea to someone with zero background. If you can't explain it to a 10-year-old or your grandmother in Amman, the idea is too weak. **"Speak Human, Not Corporate."**

L5 · ★ How Might We (HMW) — 4 Types

HMW isn't a fixed sentence — it's a **decision about which part of the problem space to explore**. It focuses thinking while keeping multiple solution directions open. Template: *"How might we help [user] achieve [job] despite [pain]?"*

1 — Efficiency	"HMW reduce waiting time?" → faster systems, optimisation
2 — Experience	"HMW reduce stress while waiting?" → emotional/perception solutions
3 — Behaviour	"HMW change when students eat?" → shift habits/patterns
4 — System	"HMW redesign how food is distributed?" → structural, biggest innovation

⚡ The HMW **type you choose determines the kind of solutions** you'll get. Too wide = chaos; too narrow = boring.

L5 · Divergence, Ideation & Communication

Divergence = expanding thinking. The first idea is the obvious one — push for **many fundamentally different** ideas. Rules: quantity over quality, no judging, different directions.

Wrong (same thinking, all variations of one idea): app with notifications, app with tracking, app with AI.
Right (different thinking): eliminate queues entirely, shift eating times, remove the need for a cafeteria.

Without an HMW: ideas are different but irrelevant. With an HMW: different **and** relevant. Change the approach, don't decorate the same idea.

Communicating ideas: filler words (umm, like, maybe) = panic signals. **The Pause** = the person who owns silence seems confident. Slow down at the insight (the "because" in your problem statement).

L6 · Validation Engine & Double Diamond

Validation Engine: everything before prototyping was in your head — now reality decides.

Insight → Hypothesis → Prototype → Test → Learn → Decide

Double Diamond — two cycles of expand-then-focus:

Discover	Expand — behaviour, tension, contradiction
Define	Focus — construct the right problem (**hardest step**)
Develop	Expand — the HMW zone, controlled exploration
Deliver	Focus — test, refine, decide

⚡ Fatal mistake: testing **HOW** (usability) before testing **WHY** (value). First confirm people want it, then refine how it works. Reality check: "if I removed the technology, would the idea still work?"

L6 · Prototype Types & Demand Signals

Type	Tests / watch for · limitation
Storyboard	Journey, sequence, logic (6–8 steps + emotion). "Does the story make sense?" — can't test demand/interaction
Role-Play	Service, conversation, support — depends on acting; not for demand
Interface (paper/digital)	Screens, user flow — watch wrong taps & hesitation; not real speed
Digital MVP	Real interaction: 1 screen / 1 flow / 1 function — watch completion & drop-off; risk of overbuilding
Landing / Fake Door	Demand — describe idea + "sign up", watch clicks/signups; tests intent, not behaviour

Outdoor test signals: "I would use this" → **ignore** (cheap talk); "Interesting, but…" → **dig** (find the real objection); "When can I get it?" → **act** (real demand). Patterns > opinions.

L7 · Feedback Extraction & Evidence Chain

Feedback isn't feedback — it's **extraction** of behaviour, friction, expectation gaps, and commitment. You didn't test ideas, you tested **assumptions**. Testing without changing anything = performance (you learned nothing).

Evidence Chain — skip any step and you're guessing:

Observation → Insight → Decision → Alignment

- **Observation** — what the user did
- **Insight** — what that behaviour means
- **Decision** — what you changed
- **Alignment** — why this better fits the user

"If you can't say this, you didn't learn": "We observed ___ · This means ___ · So we changed ___ · This aligns with the user because ___."

L7 · Feedback Methods & RAT

Silent Observation	Remove yourself — no explanation, correction or guidance. Watch hesitation, wrong actions, pauses, breakdown points.
Thinking Aloud	User narrates while doing → captures where their **mental model breaks**.
Wizard of Oz	Simulate the experience manually (you do the work behind the scenes) — does the **value** hold without the tech?
Commitment Test	Push for action/effort/continuation. If they refuse, earlier positive feedback is invalid.

RAT — Riskiest Assumption Test: identify the one assumption where "if THIS is wrong, everything collapses," and test it first. Define the success rate **and** failure threshold *before* testing. Test only with your defined persona — everyone else is noise. Never ask "Do you like it?" or "Would you use it?"

L7 · ★ Iterate / Pivot / Kill

Diagnose the **type** of problem first, then choose the action:

Signal	Diagnosis → Action
Usability Friction (HOW)	User wants value but the system blocks them → **Iterate** (fix flow/clarity)
Value Deficit (WHY)	User understands but doesn't care → **Pivot** (the problem is weak)
Mental Model Mismatch	User uses it differently than expected → **Investigate** (maybe a new opportunity)

Decision Layer: Iterate = problem real, solution weak · **Pivot** = problem wrong · **Kill** = no behavioural intent after multiple tests.

Cost of being wrong (launch stages): MVP (small, fast, iterate easily) → Pilot (controlled, slower) → Open Beta (everyone sees everything) → Polished Launch (full exposure, can't fix fast = danger). "*If you launch when you can't iterate, you're gambling, not testing.*"

⚡ EXAM STRATEGY — how to attack each question type + the traps that lose marks

🎯 How to answer each question type

- **"Classify this into SWOT / 4Ps / BMC":** ask internal-vs-external & good-vs-bad (SWOT), or which lever is being changed (4Ps). Match the keyword to the definition.
- **"Which pricing strategy?":** low price to win share = Penetration; high then drop = Skimming; based on cost = Cost-Plus; based on perceived value = Value-Based.
- **Calculation:** write the formula first, plug numbers, show working. Breakeven, margins, ROI, TAM/SAM/SOM all have fixed formulas — use them.
- **"Best single BMC change":** pick the option that fixes desirability + feasibility + viability together (usually a key partner / operations fix that also unlocks recurring revenue).
- **Design "what should they conclude":** translate the behaviour/complaint into the underlying **need**; never take the stated wish literally.
- **Iterate/Pivot/Kill:** identify HOW-problem (Iterate), WHY-problem (Pivot), or repeated no-intent (Kill).

⚠️ Most-confused distinctions

PEST vs PESTEL	PESTEL = PEST + Environmental + Legal
Fixed vs Variable	Fixed = same regardless of volume; Variable = scales with units
CapEx vs OpEx	CapEx = long-term asset (oven, used for years); OpEx = day-to-day running cost
Penetration vs Skimming	Low price to grab share vs high price first then lower
Pain vs Gain	Pain = something to avoid; Gain = a desired positive outcome
Strength vs Opportunity	Strength = internal +; Opportunity = external +
Weakness vs Threat	Weakness = internal −; Threat = external −
Profit vs Cash Flow	Can be profitable yet cash-broke (timing of payments)
TAM vs SAM vs SOM	Whole market vs serviceable slice vs obtainable short-term slice
Iterate vs Pivot	Weak solution (fix HOW) vs wrong problem (change WHY)

⚠️ Calculation traps that lose marks

- **Breakeven:** compare **total contribution** (CM×units) to fixed costs — not revenue to fixed costs.
- **Reclassification:** moving a cost between COGS and OpEx **never changes Net Profit** — only GPM moves.
- **Margins:** always divide by **revenue**; "same profit on lower revenue" = higher margin.
- **ROI vs NPM:** different denominators (invested capital vs revenue) — test each threshold separately.
- **Price ↑ in TAM/SAM/SOM:** "willing & able" % falls → SAM & SOM shrink.
- **"At least 12%":** exactly 12% **meets** the threshold (no shortfall, no headroom).
- Convert monthly→annual revenue by ×12 when asked for annual SAM revenue.

🔑 High-yield one-liners

- Innovation must create **value**, not just be invented.
- Good innovation = Desirable ∩ Feasible ∩ Viable.
- No defined segment = **no customer**.
- Observed behaviour **>** stated preference.
- Test **WHY (value)** before **HOW (usability)**.
- "I would use this" = cheap talk; "When can I get it?" = real demand.
- Beachhead needs a **shark-bite** (urgent) problem.
- **Define** is the hardest step — construct the right problem.
- If the user says YES but their system says NO → the idea dies.
- Lean = validate fast; Agile = adapt in cycles; they complement.
- Profitable ≠ cash-positive (timing of cash flows).